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Generative Artificial Intelligence
Topic: Generative Adversarial Networks (GANs)

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Abstract

This report presents an empirical study on the influence of different f-divergences and sampling strategies on Generative Adversarial Networks (GANs). We compare a baseline VanillaGAN with f-GAN variants based on Jensen–Shannon, Kullback–Leibler, and Pearson χ^2 divergences, evaluating the effects on sample quality and diversity. Several sampling procedures are explored, including normal Gaussian sampling, hard and soft truncation, and Discriminator Gradient Flow (DG$f$low) refinement. We highlight how the choice of divergence impacts gradient behavior during training and refinement, and how DG$f$low can improve sample fidelity while preserving diversity, provided the step size is carefully adapted. Our results, obtained both locally and on the course leaderboard platform, demonstrate the interplay between divergences and sampling strategies, showing that adaptive refinement can enhance GAN performance without modifying the trained generator.

1 Introduction

Generative Adversarial Networks (GANs) [2] have emerged as a powerful class of models for generating realistic data across various domains. Despite their success, training GANs remains challenging due to issues such as mode collapse, instability, and sensitivity to hyperparameters. Evaluating the performance of GANs is also non-trivial, requiring metrics that capture both the quality and diversity of generated samples.
In this project, we conduct a systematic comparison of multiple GAN variants on the MNIST dataset [6], focusing on a fixed generator architecture. Our goal is to understand how different training objectives, sampling strategies, and post-hoc refinement methods affect the generation performance. In particular, we study:

- The baseline GAN using the standard Jensen-Shannon divergence.
- f-GAN [8] variants based on Kullback–Leibler (KL), Pearson, and Jensen–Shannon (JS) divergences, which modify the discriminator loss to encourage different properties in the generated samples.
- Advanced sampling techniques including hard [1] and soft [4] truncation of the latent space.
- Discriminator Gradient Flow (DG$f$low) refinement, which leverages the discriminator to iteratively improve generated images.

We evaluate the models using metrics that quantify both sample quality and diversity, namely FID (Fréchet Inception Distance) [3], Precision, and Recall [5]. This study provides insights into the trade-offs between different training objectives and sampling methods, and highlights best practices for training GANs on simple image datasets like MNIST.

2 Related Work

Generative Adversarial Networks (GANs) are based on an adversarial training framework where a generator learns to produce samples that mimic a real data distribution, while a discriminator attempts to distinguish real samples from generated ones. The training objective corresponds to a minimax game that approximates the JS divergence between the real and generated distributions. Although this approach has achieved strong results in generative modeling, it is well known that GAN training can be unstable and may suffer from issues such as mode collapse.
To address these limitations, the f-GAN framework generalizes GAN training by minimizing a broader class of statistical divergences known as f-divergences. Instead of restricting the objective to the JS divergence, this formulation allows the use of alternative divergences such as the KL divergence or the Pearson χ^2 divergence. The framework relies on a variational formulation that enables the estimation and minimization of these divergences through adversarial training, providing a unified perspective on different GAN objectives.

Another direction of research focuses on improving generated samples after the generation process. Discriminator Gradient Flow is a refinement technique that uses the gradient of the discriminator with respect to the input samples. Starting from a generated image, the method iteratively updates the sample by following the discriminator gradient in order to move it toward regions that the discriminator considers more realistic. This post-processing step aims to improve the quality of generated samples without modifying the generator architecture.
In addition to modifying training objectives or refining samples, several works have investigated alternative sampling strategies in the latent space. Truncation methods restrict the latent variables to a subset of the latent distribution with higher probability density. Such approaches can improve the visual quality of generated samples but may reduce the diversity of the outputs.

These different approaches provide complementary strategies for improving GAN performance. In this project, we experimentally study these techniques under a common generator architecture and evaluate their impact on the generation of MNIST images.

3 Evaluation Metrics and Sampling Strategies

Before studying the different GAN variants, several evaluation metrics and sampling strategies were implemented in order to provide a consistent framework for comparing generative models. Particularly in the beginning, we implemented three evaluation metrics – FID, Precision, and Recall – as well as two latent space sampling techniques: Hard Truncation and Soft Truncation.

Fréchet Inception Distance (FID). The Fréchet Inception Distance (FID) is a commonly used metric for evaluating the quality of generated images. It measures the distance between the feature distributions of real and generated images.
Let μ_r, Σ_r denote the mean and covariance of the feature representations of real images, and μ_g, Σ_g the corresponding statistics for generated images. These statistics are computed from features extracted by a neural network. The FID score is defined as

$$FID = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right).$$

A lower FID score indicates that the generated samples are closer to the real data distribution, capturing both the similarity of feature means and feature covariances.

Precision and Recall. While FID measures overall distribution similarity, it does not explicitly distinguish between sample quality and sample diversity. Precision and Recall were introduced to address this limitation.
Precision measures the proportion of generated samples that lie within the support of the real data distribution. If $\mathcal{M}_r$ denotes the manifold approximating the real data and $\mathcal{M}_g$ the manifold of generated samples, precision can be interpreted as

$$\text{Precision} = \frac{|\mathcal{M}_g \cap \mathcal{M}_r|}{|\mathcal{M}_g|}.$$

A high precision score indicates that generated samples are realistic.
Recall measures the extent to which the generator captures the diversity of the real dataset and can be expressed as

$$\text{Recall} = \frac{|\mathcal{M}_g \cap \mathcal{M}_r|}{|\mathcal{M}_r|}.$$

A high recall value indicates that the generator successfully covers the different modes of the real data distribution.
For the implementation of Precision and Recall, we followed the methodology proposed in [5]. Since the original formulation relies on feature representations of images, a dedicated feature extractor was trained on the MNIST dataset in order to obtain a meaningful embedding space. The network was trained as a classification model and intermediate representations were used as feature vectors describing each image.
Once trained, this feature extractor was used to map both real and generated samples into a common feature space. Precision and Recall were then computed by comparing the geometric structure of the two distributions in this embedding space, allowing us to estimate both the realism of generated samples and the diversity captured by the generator.

Latent Space Truncation. In addition to evaluation metrics, truncation techniques were implemented to modify how latent vectors are sampled before being passed to the generator. These methods restrict the latent distribution to regions with higher probability density, which can improve sample quality but may reduce diversity. The following methods were implemented to address this:

1. **Hard Truncation:** Hard truncation restricts the latent variables to lie within a fixed interval. If the latent variable z follows a standard normal distribution $z \sim \mathcal{N}(0,1)$, Hard Truncation imposes the constraint

$$z' = \text{clip}(z, -\tau, \tau)$$

where τ is the truncation threshold. Values outside the interval $[-\tau, \tau]$ are clipped to the boundary, reducing extreme latent values that may lead to unrealistic generations.

2. **Soft Truncation:** Soft truncation modifies the latent variable more smoothly by scaling large values instead of abruptly clipping them. One possible formulation is

$$z' = z \cdot \min\left(1, \frac{\tau}{|z|}\right).$$

Unlike Hard Truncation, this transformation preserves continuity in the latent space while still reducing the magnitude of extreme latent values, often providing a better balance between sample quality and diversity.

4 f-GAN and f-Divergence Generalization

The original GAN objective can be interpreted as minimizing the Jensen–Shannon (JS) divergence between the real data distribution p_{data} and the generator distribution p_g. Although this formulation has proven successful in many applications, it restricts the adversarial training objective to a single divergence measure.
The f-GAN framework generalizes this idea by allowing the minimization of a broader class of statistical divergences known as f-divergences. Given two probability distributions p and q, an f-divergence is defined as

$$D_f(p\|q) = \int q(x)\, f\left(\frac{p(x)}{q(x)}\right) dx,$$

where $f : \mathbb{R}_+ \to \mathbb{R}$ is a convex function satisfying $f(1) = 0$. Different choices of f correspond to different divergence measures. For instance, appropriate choices recover the Kullback–Leibler divergence, the Jensen–Shannon divergence, or the Pearson χ^2 divergence.
A key contribution of the paper is to show that these divergences admit a variational representation based on convex duality. In particular, using the Fenchel conjugate f^* of the function f, the divergence can be written as

$$D_f(p\|q) = \sup_T \left(\mathbb{E}_{x\sim p}[T(x)] - \mathbb{E}_{x\sim q}[f^*(T(x))]\right),$$

where the supremum is taken over a class of measurable functions T. In practice, this function is parameterized by a neural network that plays a role similar to the discriminator in standard GANs.
One advantage of this formulation is that different divergences emphasize different discrepancies between distributions. For example, the KL divergence tends to penalize missing modes more strongly, while the Pearson χ^2 divergence behaves differently with respect to density ratios. By selecting an appropriate divergence, it may therefore be possible to influence the behavior and stability of the training process.
In the original paper, several divergences were experimentally evaluated and compared using the Kernel Density Estimation (KDE) score, a metric that estimates the likelihood of generated samples under a non-parametric density model fitted on the real data. According to the results reported in Table 4 of the paper, the divergences that achieved the best KDE scores were the Jensen–Shannon divergence, the Kullback–Leibler divergence, and the Pearson χ^2 divergence.
The three divergences we implemented are summarized in the following table:

Divergence	**Generator** $f(u)$
Jensen–Shannon (JS)	$u \log u - (u+1) \log \frac{u+1}{2}$
Kullback–Leibler (KL)	$u \log u$
Pearson χ^2	$(u-1)^2$

Here, $u = \frac{p(x)}{q(x)}$ represents the ratio between the densities of the real data and the generator at point x.
This flexibility allows the practitioner to adapt the training objective to the dataset and generator architecture, potentially improving stability, reducing mode collapse, or emphasizing sample fidelity depending on the chosen divergence.
By focusing on these three divergences, we maintain a manageable experimental setup while exploring how different f-divergence choices influence the generator's performance.

5 Refining Generative Models via Discriminator Gradient Flow (DG$f$low)

DG$f$low introduces a post-hoc refinement technique that iteratively updates generated samples in latent space using the gradient of a discriminator, moving them toward regions deemed more realistic. Unlike standard sampling from the latent space, which draws latent vectors from a predefined distribution (e.g., standard normal), DG$f$low uses the discriminator to refine each sample individually. This sample-specific guidance allows corrections beyond what is achievable by sampling alone or by truncation-based methods.
DG$f$low refines a latent vector z according to

$$z_{t+1} = z_t - \eta \nabla_z f'(r(x)), \quad x = G(z_t),$$

where $r(x) = p_{data}(x)/p_g(x)$ is the density ratio at x, f' is the derivative of the chosen f-divergence, η is the step size, and t indexes the refinement iteration. The update iteratively moves each sample toward regions with higher probability under the real data distribution while respecting the geometry induced by the chosen divergence.

Advantages over Sampling and Truncation.

- **Standard sampling:** Draws latent vectors from a fixed distribution (e.g., normal), producing unrefined samples that may lie in low-density or unrealistic regions.
- **Hard/Soft truncation:** Constrains latent vectors globally to reduce extreme values, improving quality but potentially reducing diversity.
- **DG$f$low:** Applies sample-specific corrections using discriminator gradients, improving realism while maintaining diversity and going beyond what global truncation can achieve.

In the DG$f$low paper, KL and JS divergences were used. Pearson χ^2 was avoided due to instability in the refinement process. The derivatives used are summarized below:

Divergence	$f'(r)$
KL	$\log r + 1$
JS	$\log \frac{2r}{r+1}$

Pearson χ^2, with $f(r) = (r-1)^2$ and derivative $f'(r) = 2(r-1)$, can lead to several problems in DG$f$low. In fact, the authors of the DG$f$low paper did not experiment with this divergence due to these stability issues:

1. **Exploding gradients:** When r is very large (samples far from the real data manifold), the linear derivative produces huge updates in latent space.
2. **Abrupt sign changes:** For $r < 1$ (samples already close to the real data), the derivative switches sign sharply, causing oscillations.
3. **Sensitivity to outliers:** Large deviations in r amplify latent updates excessively, destroying the structure of generated samples.
4. **Comparison with KL/JS:** Logarithmic derivatives like $\log r + 1$ (KL) or $\log \frac{2r}{r+1}$ (JS) provide smoother, bounded updates that guide samples gently toward higher-density regions.

DG$f$low complements both standard sampling and latent truncation by refining individual samples using discriminator feedback. Using divergences with controlled gradients ensures improved visual fidelity without compromising diversity, while poorly-behaved divergences like Pearson may produce unstable refinements.

6 Implementation and Practical Considerations

Several implementation choices were made to ensure stable training and to maintain consistent experimental conditions across the different divergences and sampling strategies explored in this work.

Discriminator Architectures. For the baseline GAN (VanillaGAN), a feed-forward discriminator with Leaky ReLU activations and dropout was used for stabilization. The architecture is as follows: input layer matching the data dimension (e.g., 784 for MNIST), hidden layers $1024 \to 512 \to 256$ with Leaky ReLU activations, dropout (p=0.3) after each hidden layer to reduce overfitting and improve training stability, and a single output neuron with sigmoid activation. This discriminator was used only for VanillaGAN.
For f-GAN training, a slightly different discriminator was implemented to model the variational function associated with f-divergences. The network follows the same layer structure $1024 \to 512 \to 256 \to 1$, with spectral normalization [7] applied to each linear layer. Spectral normalization constrains the spectral norm (i.e., the largest singular value) of the weight matrices, limiting the discriminator's ability to amplify input perturbations. This improves training stability in adversarial settings by preventing the discriminator from becoming overly powerful relative to the generator. Leaky ReLU activations are used in the hidden layers, and the sigmoid activation at the output is removed. This is because, in f-GAN, the discriminator does not model a probability but rather a real-valued variational function

used to estimate an f-divergence. The variational objective requires unconstrained logits; applying a sigmoid would restrict the output range, distort the divergence estimator, and introduce gradient saturation. Stability is therefore ensured through spectral normalization rather than probabilistic output constraints. This design follows practical recommendations from the f-GAN paper, including monitoring real/fake statistics, using Adam optimizers, and gradient clipping for large-scale datasets.

Divergence-Specific Implementations. The f-GAN implementation supports KL, JS, and Pearson χ^2 divergences. Activation functions $g_f(v)$ are identity for KL and Pearson, and a numerically stable log-based activation for JS: $g_f(v) = \log(2) - \text{softplus}(-v)$. Fenchel conjugates $f^*(t)$ are defined as: KL: $f^*(v) = \exp(v-1)$, Pearson: $f^*(v) = 0.25v^2 + v$, JS: $f^*(t) = -\log(2 - \exp(t))$, implemented with clamping to ensure numerical stability. Generator loss updates follow standard f-GAN rules: for KL and Pearson maximize $\mathbb{E}[g_f(T(\text{fake}))]$, for JS a stable softplus form is used. These implementations respect the variational formulation of f-divergences, linking discriminator outputs to density ratios and derivatives.

DG$f$low Sampling and Refinement. DG$f$low allows post-hoc refinement of latent vectors z. The step size η controls the magnitude of updates applied in latent space. In practice, $\eta = 0.1$ was used for KL and JS divergences, following the default value suggested in the original paper. However, the Pearson χ^2 divergence requires a significantly smaller step size in order to maintain stable updates. This is due to the linear growth of the derivative associated with the Pearson divergence, which can produce excessively large gradients when the density ratio becomes large. As discussed previously, this behavior may cause latent vectors to diverge during the refinement process. Empirical experiments confirmed that DG$f$low must therefore be adapted for Pearson by reducing the step size substantially in order to avoid unstable updates and obtain meaningful refinements.
The number of refinement steps is typically set to 25, with Gaussian noise ($\gamma = 0.01$) added at each iteration to preserve diversity. Updates are applied iteratively using discriminator gradients while freezing generator weights. DG$f$low complements standard sampling strategies: normal Gaussian sampling, hard truncation (clipping beyond a threshold), soft truncation (smooth bias toward the origin), and DG$f$low itself (sample-specific refinement using divergence gradients). This approach allows sample-specific corrections without globally constraining the latent distribution.

Training Hyperparameters. Batch size: 64 for training, 2048 for generation. Number of epochs: 600. Learning rates: generator 2×10^{-4} for all divergences; discriminator: 1×10^{-4} for JS (slower learning to maintain balance), 2×10^{-4} for KL and Pearson. Optimizer: Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$). Gradient clipping prevents exploding gradients, especially for KL and Pearson. Leaky ReLU activations in hidden layers, optional sigmoid on output.

Specific Considerations for JS Divergence. Numerical stability is critical: $f^*(t) = -\log(2 - \exp(t))$ may produce NaN if $t \to \log(2)$. The activation $g_f(v)$ ensures outputs remain below $\log(2)$, protecting the computation. Using JS in f-GAN closely reproduces VanillaGAN behavior, since BCELoss implicitly minimizes Jensen-Shannon divergence. JS also benefits from DG$f$low, improving refinement stability compared to Pearson.
Overall, the implementation integrates discriminator design (including spectral normalization for f-GAN), divergence-specific updates, DG$f$low refinement, and tuned hyperparameters. This reproduces the f-GAN methodology faithfully and extends it with sample-specific latent refinement for improved visual fidelity while preserving diversity.

7 Results

The results presented in this section correspond to the metrics obtained both from experiments performed on our local machines and from submissions evaluated on the course leaderboard platform[1]. These metrics provide a quantitative comparison between the different models and sampling strategies considered in this project.

Model	Sampling	FID	Precision	Recall
VanillaGAN	Normal	0.0208	0.7338	0.8256
	Soft Truncation	0.0664	0.8264	0.6528
	Hard Truncation	0.4905	0.8938	0.2015
	DG$f$low	0.0207	0.7460	0.8270
f-GAN JS	Normal	0.0059	0.8025	0.7935
	Soft Truncation	0.0384	0.8730	0.6227
	Hard Truncation	0.2592	0.8780	0.2496
	DG$f$low	0.0056	0.8417	0.7504
f-GAN KL	Normal	0.0080	0.7524	0.8211
	Soft Truncation	0.0283	0.8316	0.6635
	Hard Truncation	0.2399	0.8306	0.2903
	DG$f$low	0.0106	0.7913	0.7328
f-GAN Pearson	Normal	0.0059	0.7909	0.7868
	Soft Truncation	0.0264	0.8537	0.6497
	Hard Truncation	0.2184	0.8663	0.2385
	DG$f$low	0.0060	0.8127	0.7879

Since preliminary experiments performed locally consistently showed that soft and hard truncation never outperformed normal sampling or DG$f$low, these strategies were not submitted to the leaderboard platform. Also, only the results obtained with f-GAN models are reported for the official submissions.

Model	Sampling	FID	Precision	Recall
f-GAN JS	Normal	8.51	0.86	0.64
	DG$f$low	7.38	0.87	0.62
f-GAN KL	Normal	11.62	0.84	0.66
	DG$f$low	10.56	0.84	0.62
f-GAN Pearson	Normal	9.78	0.84	0.66
	DG$f$low	9.3	0.85	0.65

[1] Official test platform: `https://www.lamsade.dauphine.fr/~testplatform/iasdapp/results/last.html`

Interpretation of Local Results. The first table provides a detailed comparison across all models and sampling strategies. Several observations can be made. First, the use of hard truncation consistently leads to a strong degradation in recall while increasing precision. This behavior is expected, as truncation restricts the latent space and therefore reduces diversity in generated samples. Soft truncation shows a milder version of the same trade-off: precision tends to increase while recall decreases.

Normal sampling generally provides a balanced trade-off between precision and recall. However, DG$f$low often slightly improves the quality of generated samples, particularly in terms of FID and precision, while maintaining competitive recall values. This is consistent with the intuition behind DG$f$low, which refines individual latent vectors using discriminator gradients in order to move generated samples closer to high-density regions of the data distribution.

Comparing divergences, the f-GAN variants generally outperform the VanillaGAN baseline in terms of FID. In particular, the Jensen-Shannon and Pearson divergences achieve the lowest FID values under normal sampling and DG$f$low. This suggests that the variational formulation of f-divergences can provide a more stable training objective compared to the standard GAN formulation.

Interpretation of Leaderboard Results. The second table reports the results obtained on the course evaluation platform. These experiments focus only on normal sampling and DG$f$low for the f-GAN models. The results confirm the trends observed locally: DG$f$low slightly improves the FID score for both the JS and KL divergences, while maintaining similar precision and recall values. For the Pearson χ^2 divergence, DG$f$low also provides a small improvement in FID and precision while keeping recall at a comparable level.

Although the improvements remain moderate, they indicate that latent refinement through DG$f$low can enhance sample quality without significantly sacrificing diversity.

8 Conclusion

This project explored the impact of different f-divergences and sampling strategies on the quality and diversity of samples generated by GAN models. In particular, we compared a baseline VanillaGAN with several f-GAN variants based on Jensen–Shannon, Kullback–Leibler, and Pearson χ^2 divergences, and evaluated how different sampling procedures influence the resulting generative performance.

Our experiments highlighted several important observations. First, the choice of divergence significantly affects the behavior of the discriminator and therefore the gradients used during training and refinement. Divergences such as JS and KL provide smooth logarithmic gradients that guide the generator toward high-density regions of the data distribution, leading to stable training and strong sample quality. In contrast, the Pearson χ^2 divergence may produce large gradients when the density ratio becomes large, requiring additional care in the refinement process.

Second, sampling strategies have a strong influence on the trade-off between precision and recall. Hard truncation tends to improve precision by concentrating samples in high-density regions, but it significantly reduces recall by limiting diversity. Soft truncation provides a milder bias toward the center of the latent distribution but still degrades diversity compared to standard sampling. DG$f$low offers a different perspective by refining individual samples using discriminator feedback rather than globally constraining the latent distribution.

The experimental results confirm that DG$f$low can improve certain metrics depending on the divergence used, particularly when the gradients remain well behaved. However, the effectiveness of this refinement step depends on the properties of the divergence and requires careful tuning of the update step size to avoid instability. This is especially important for divergences such as Pearson χ^2, where overly large updates may destabilize the latent refinement process.

Overall, this study illustrates how both the choice of divergence and the sampling strategy interact to shape the final performance of generative models. While standard sampling already produces strong results for several configurations, post-hoc refinement techniques such as DG$f$low provide an additional mechanism to improve sample quality without modifying the trained generator. Future work could explore more adaptive refinement strategies, alternative divergences, or scaling these approaches to larger and more complex datasets.

References

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Appendix: Generated Images

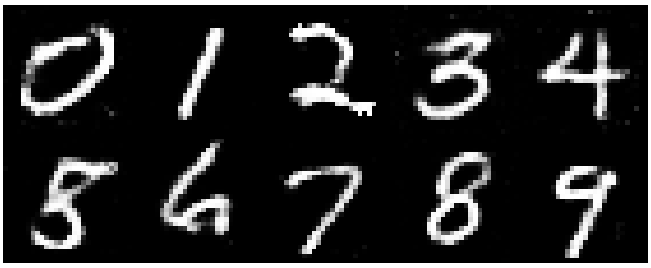

Figure 1: VanillaGAN – Normal

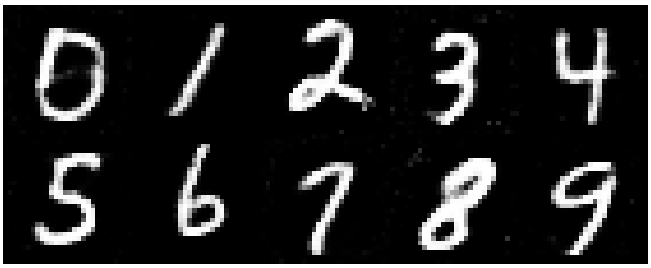

Figure 2: VanillaGAN – Soft Truncation

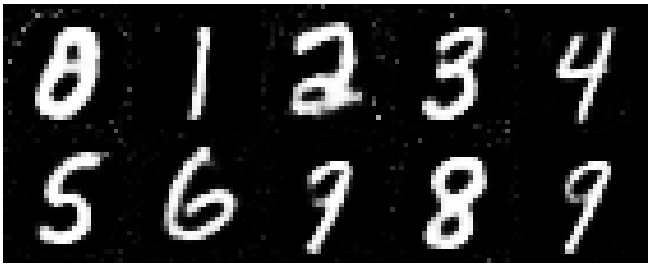

Figure 3: VanillaGAN – Hard Truncation

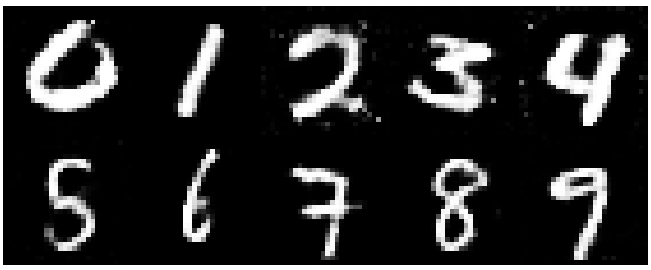

Figure 4: VanillaGAN – DG$f$low

Figure 5: f-GAN JS – Normal

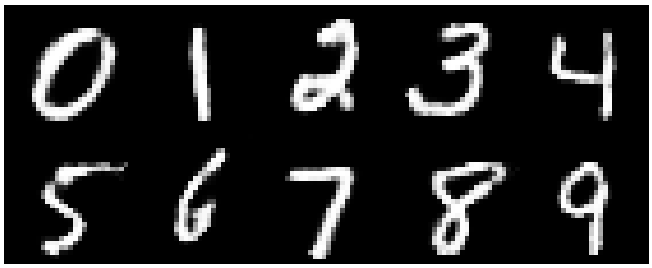

Figure 6: f-GAN JS – Soft Truncation

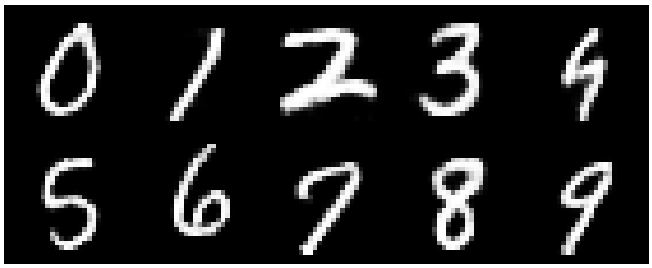

Figure 7: f-GAN JS – Hard Truncation

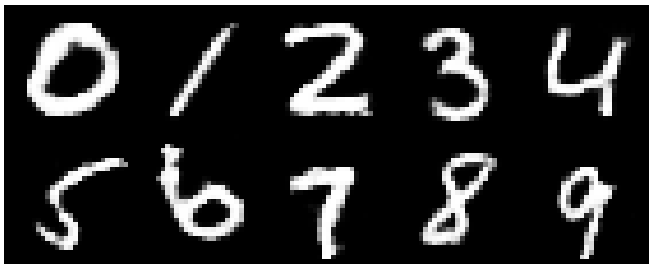

Figure 8: f-GAN JS – DG$f$low

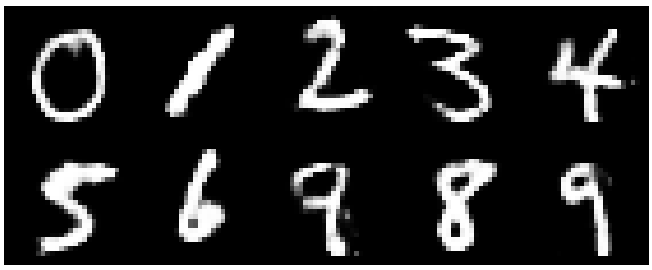

Figure 9: f-GAN KL – Normal

Figure 10: f-GAN KL – Soft Truncation

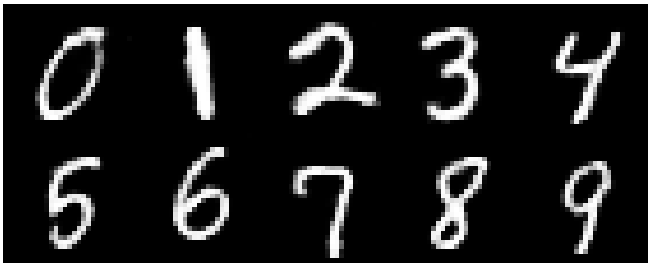

Figure 11: f-GAN KL – Hard Truncation

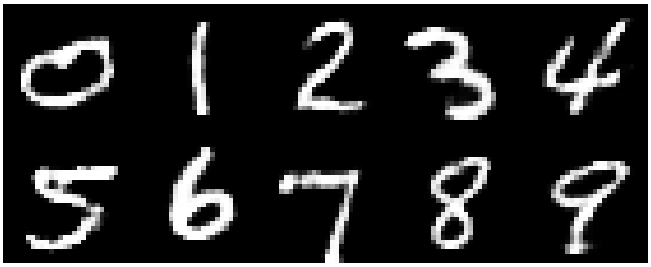

Figure 12: f-GAN KL – DG$f$low

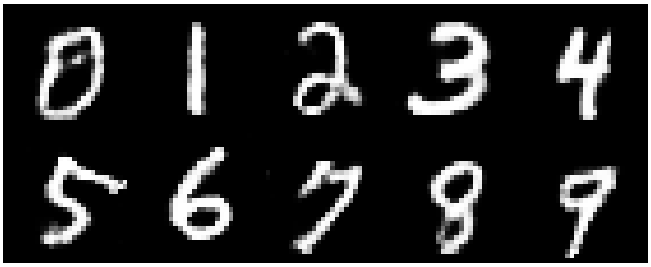

Figure 13: f-GAN Pearson – Normal

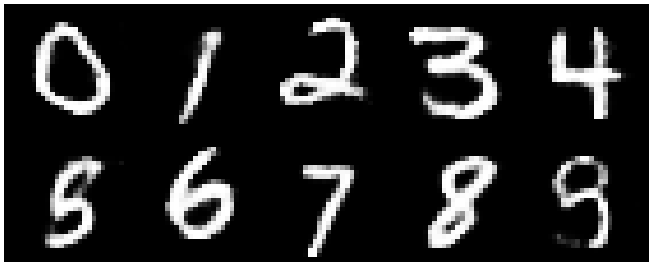

Figure 14: f-GAN Pearson – Soft Truncation

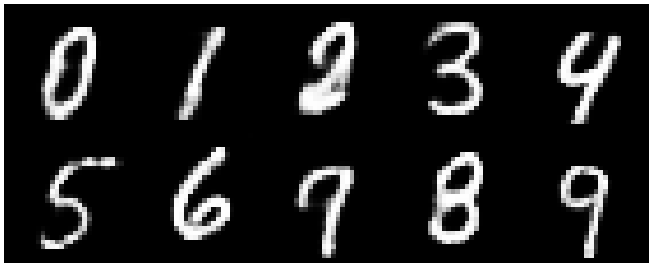

Figure 15: f-GAN Pearson – Hard Truncation

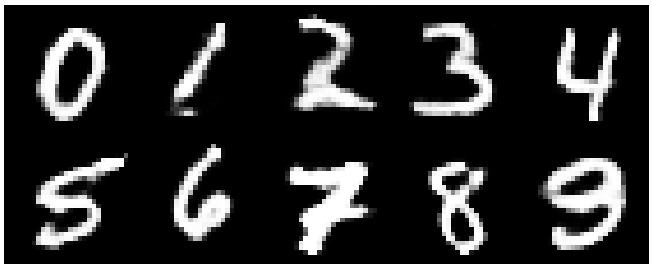

Figure 16: f-GAN Pearson – DG$f$low